Agentic Data Scientist Report

> **Dataset:** `../../tests/data/prediction/historical_daily_2025.csv` **Target:** `None`

1. Dataset Profile

Property	Value
Rows	**365**
Initial columns	**12**
Final columns	**16**
Task	N/A
Learning	Unsupervised
Imbalance ratio	N/A

1.1 Initial Feature Types

- **Boolean** (0): None
- **Numeric** (24): `weather_code`, `daylight_duration`, `shortwave_radiation_sum`, `precipitation_sum`, `snowfall_sum`, `precipitation_hours`, `temperature_2m_max`, `temperature_2m_min`, `date_utc_year`, `date_utc_month`, `date_utc_day`, `date_utc_dayofweek`, `sunrise_year`, `sunrise_month`, `sunrise_day`, `sunrise_dayofweek`, `sunrise_hour`, `sunrise_minute`, `sunset_year`, `sunset_month`, `sunset_day`, `sunset_dayofweek`, `sunset_hour`, `sunset_minute`
- **Date** (0): None
- **Text** (0): None
- **Categorical** (0): None

1.2 Final Feature Types

- **One-Hot Encoded** (0): None
- **Embedded** (0): None
- **Scaled** (16): `weather_code`, `daylight_duration`, `shortwave_radiation_sum`, `precipitation_sum`, `snowfall_sum`, `precipitation_hours`, `temperature_2m_max`, `temperature_2m_min`, `date_utc_year`, `date_utc_month`, `date_utc_day`, `date_utc_dayofweek`, `sunrise_hour`, `sunrise_minute`, `sunset_hour`, `sunset_minute`
- **Unchanged** (0): None

2. Data Quality

Check	Result
Missing values	No missing values found.
Systematic missingness	No columns with missing data to assess
Skewed columns	`shortwave_radiation_sum`, `precipitation_sum`, `snowfall_sum`, `precipitation_hours`
Duplicates	0 duplicate row(s), 8 duplicate column(s) found.
Outliers	IQR-based removal (training only): `precipitation_sum`, `rain_sum`, `snowfall_sum`

3. Statistical Properties

Feature	Mean	Median	Std	Min	Max	Q1	Q3	Skewness
weather_code	39.1452	51.0	26.5907	0.0	75.0	3.0	61.0	-0.5428
daylight_duration	44302.1802	44470.41	12313.4471	26013.016	62307.086	32640.715	56042.562	-0.0165
shortwave_radiation_sum	10.6207	9.56	7.9644	0.52	29.04	3.05	17.02	0.4587

Feature	Mean	Median	Std	Min	Max	Q1	Q3	Skewness
precipitation_sum	3.5493	0.9	5.8132	0.0	35.4	0.0	4.6	2.5844
rain_sum	3.4405	0.7	5.7673	0.0	35.4	0.0	4.4	2.6597
snowfall_sum	0.0761	0.0	0.619	0.0	10.78	0.0	0.0	14.8347
precipitation_hours	6.3973	4.0	6.7375	0.0	24.0	0.0	11.0	0.8054
temperature_2m_max	12.1025	12.7	5.5278	-1.65	26.3	8.05	16.2	-0.0474
temperature_2m_min	5.7314	5.9	4.802	-6.9	16.45	2.1	9.65	-0.1194

Property	Details
Normality	**Normal:** None **Non-normal:** `weather_code`, `daylight_duration`, `shortwave_radiation_sum`, `precipitation_sum`, `snowfall_sum`, `precipitation_hours`, `temperature_2m_max`, `temperature_2m_min`, `date_utc_year`, `date_utc_month`, `date_utc_day`, `date_utc_dayofweek`, `sunrise_hour`, `sunrise_minute`, `sunset_hour`, `sunset_minute`
Scaling applied	**StandardScaler:** None **MinMaxScaler:** `weather_code`, `daylight_duration`, `shortwave_radiation_sum`, `precipitation_sum`, `snowfall_sum`, `precipitation_hours`, `temperature_2m_max`, `temperature_2m_min`, `date_utc_year`, `date_utc_month`, `date_utc_day`, `date_utc_dayofweek`, `sunrise_hour`, `sunrise_minute`, `sunset_hour`, `sunset_minute`
Multicollinearity	**Pairs (
Feature importance	N/A
Train/Test split	Train: 255 rows, Test: 110 rows (70/30 split).

4. Preprocessing

Step	Details
Binary-encoded variables	None
One-hot encoded variables	None
Label-encoded target	Not applicable (numeric target)
Log-transformed variables	`shortwave_radiation_sum`, `precipitation_sum`, `snowfall_sum`, `precipitation_hours`
Standard scaling	None
MinMax scaling	`weather_code`, `daylight_duration`, `shortwave_radiation_sum`, `precipitation_sum`, `snowfall_sum`, `precipitation_hours`, `temperature_2m_max`, `temperature_2m_min`, `date_utc_year`, `date_utc_month`, `date_utc_day`, `date_utc_dayofweek`, `sunrise_hour`, `sunrise_minute`, `sunset_hour`, `sunset_minute`

5. Modelling Concerns

1. Small dataset (<1000 rows): prefer simpler models / guard against overfitting.
2. The agent dropped 1 multicollinear column(s): ['rain_sum']. The columns that are kept favour linear predictive signal. This is an issue if dropped features still carry non-linear information useful to tree-based or kernel models.

3. Removed 27 outlier row(s). This IQR-based removal can be aggressive on skewed distributions. Furthemore, outliers remain in the test set, so train/test distributions may diverge.

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6. Processing Notes

1. Columns with IQR outliers: ['precipitation_sum', 'rain_sum', 'snowfall_sum']
2. Columns with skew (|skew| > 0.5): ['weather_code', 'precipitation_sum', 'rain_sum', 'snowfall_sum', 'precipitation_hours']
3. No numeric target, we therefore dropped 'rain_sum' which has lower variance than 'precipitation_sum' (lower variance: 33.2617 vs 33.7933).
4. Converted datetime column 'date_utc' to numeric components.
5. Converted datetime column 'sunrise' to numeric components.
6. Converted datetime column 'sunset' to numeric components.
7. Datetime columns converted to numeric components.
8. Dropped duplicate column(s): ['sunrise_year', 'sunset_year', 'sunrise_month', 'sunset_month', 'sunrise_day', 'sunset_day', 'sunrise_dayofweek', 'sunset_dayofweek'].
9. Removed 27 outlier row(s) from training data (IQR fences) in columns: ['precipitation_sum', 'rain_sum', 'snowfall_sum'].

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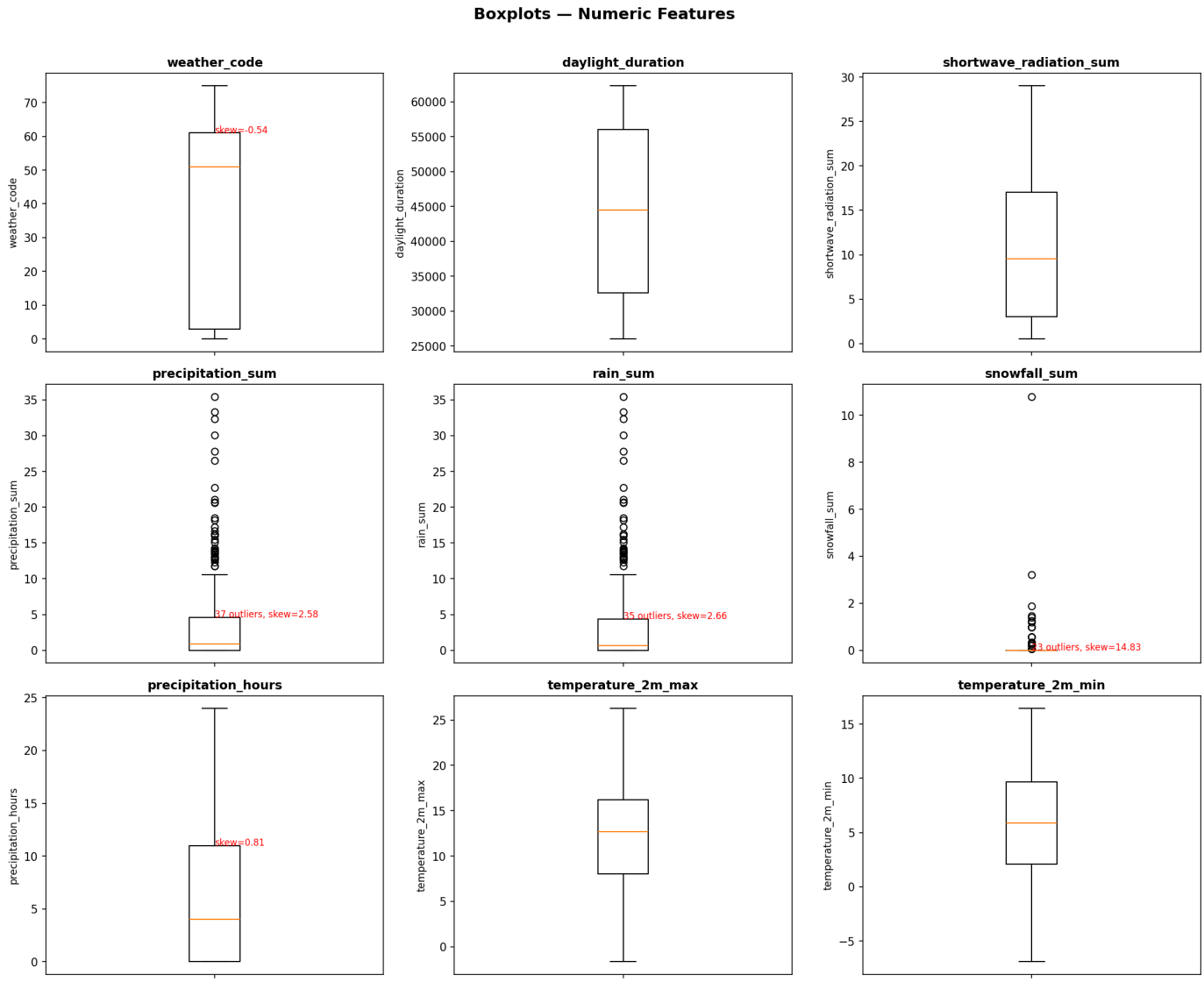
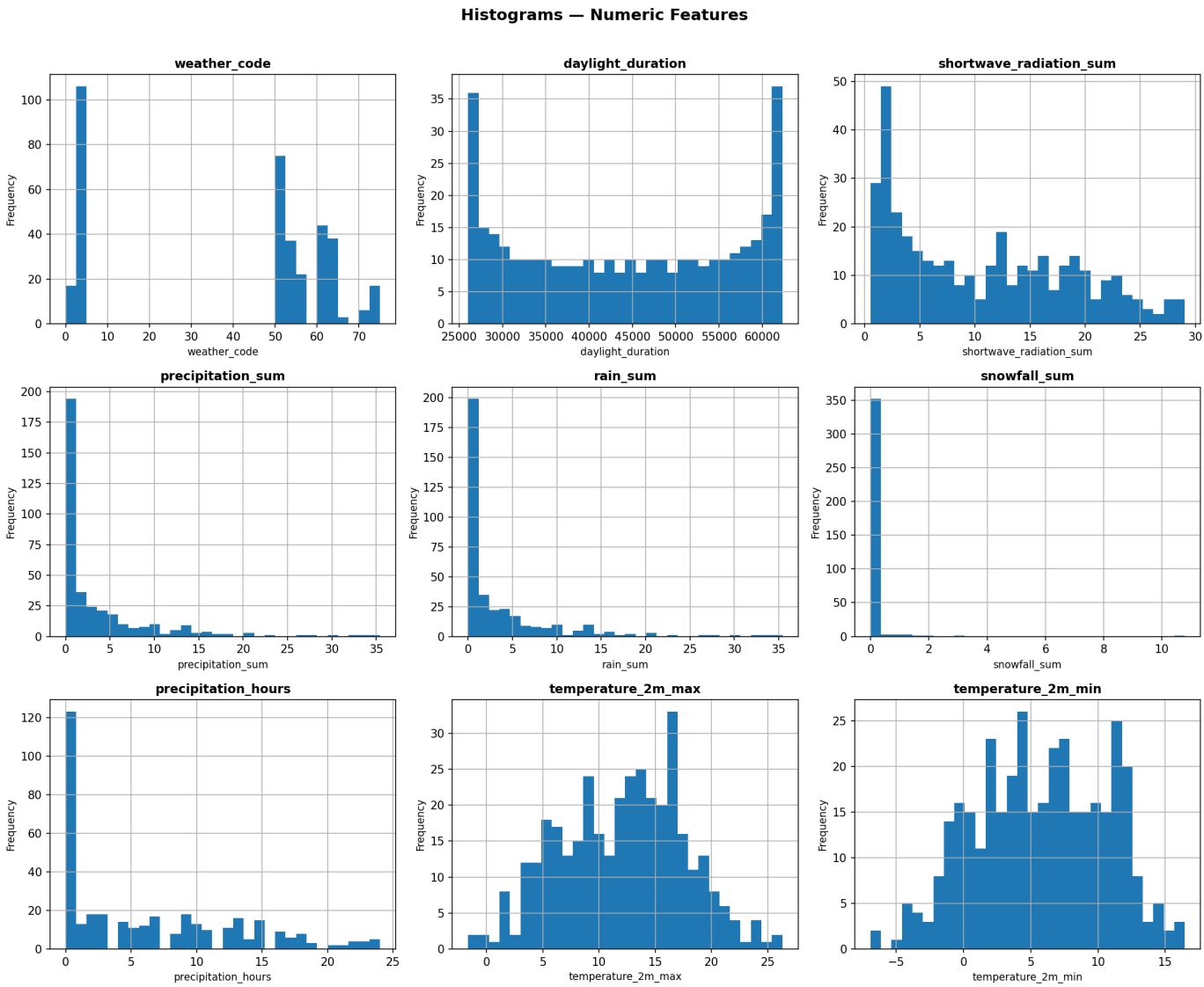
7. EDA Visualisations

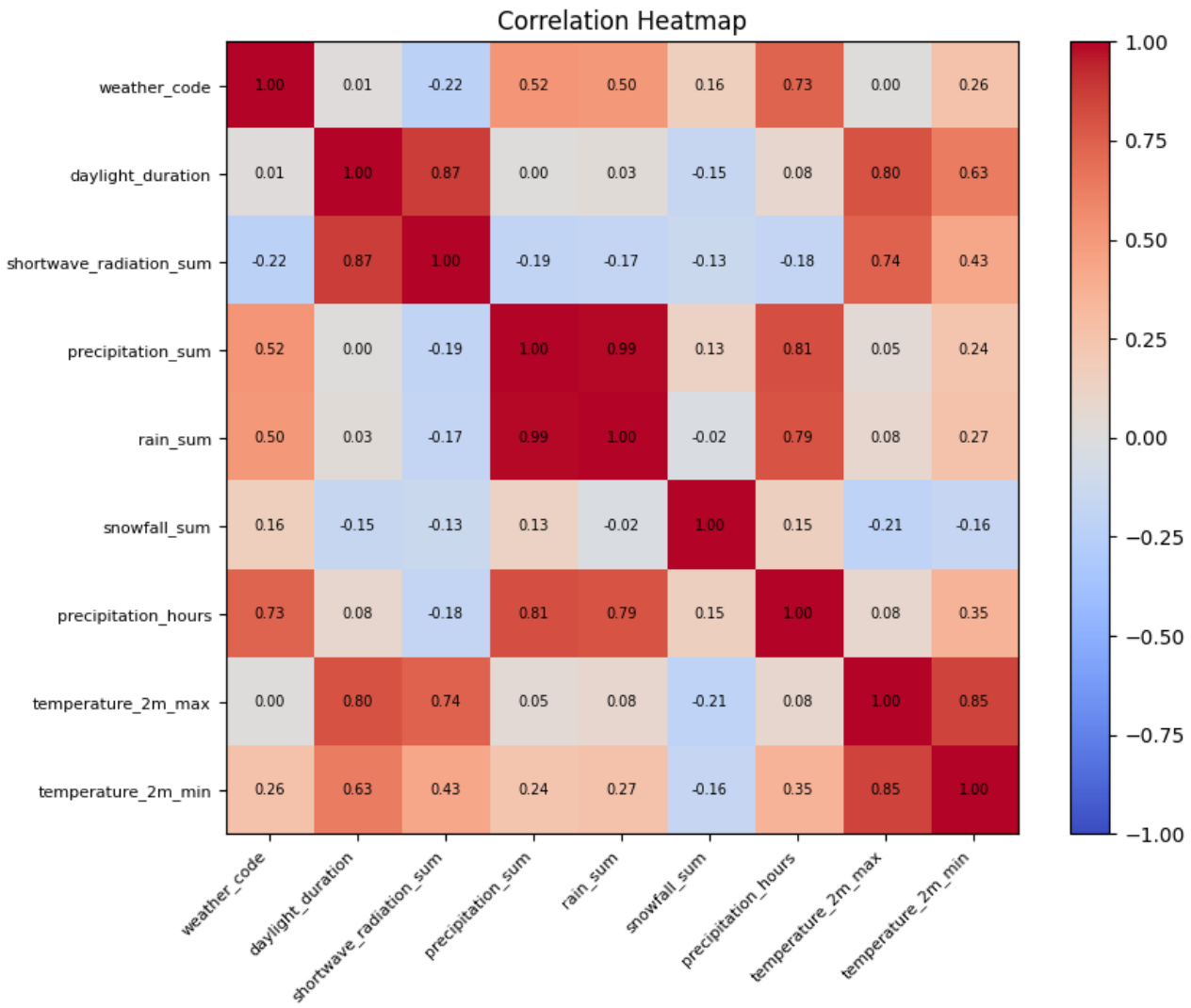
Boxplots — Numeric Features



Histograms — Numeric Features



Correlation Heatmap


Correlation Heatmap
weather_code
daylight_duration
shortwave_radiation_sum
precipitation_sum
rain_sum
snowfall_sum
precipitation_hours
temperature_2m_max
temperature_2m_min
1.00 0.01 -0.22 0.52 0.50 0.16 0.73 0.00 0.26
0.01 1.00 0.87 0.00 0.03 -0.15 0.08 0.80 0.63
-0.22 0.87 1.00 -0.19 -0.17 -0.13 -0.18 0.74 0.43
0.52 0.00 -0.19 1.00 0.99 0.13 0.81 0.05 0.24
0.50 0.03 -0.17 0.99 1.00 -0.02 0.79 0.08 0.27
0.16 -0.15 -0.13 0.13 -0.02 1.00 0.15 -0.21 -0.16
0.73 0.08 -0.18 0.81 0.79 0.15 1.00 0.08 0.35
0.00 0.80 0.74 0.05 0.08 -0.21 0.08 1.00 0.85
0.26 0.63 0.43 0.24 0.27 -0.16 0.35 0.85 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00